

PAPER159 — 13th Resonance Term: Morris-Thorne Wormhole in MUGE ($a_{\text{worm}} = f_{\text{worm}} \cdot E_{\text{vac}} / (b^2 + r^2)$)

Session: 47 | Date: March 13, 2026 | Thread: 7f9068 | Domain: §2.3

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Abstract

This paper documents the addition of a **13th term** to the MUGE Resonance Master Equation, extending PAPER146 (12-term) by incorporating a Morris-Thorne traversable wormhole gravitational contribution: $a_{\text{worm}}(r) = f_{\text{worm}} \cdot E_{\text{vac,neb}} / (b^2 + r^2)$. This term was isolated from C++ execution in Grok thread 7f9068 and is **distinct** from the Morris-Thorne geodesic metric treatment in PAPER153. PAPER_153 concerns the wormhole spacetime metric; this paper concerns the **gravitational acceleration** contribution to MUGE from the wormhole throat vacuum energy.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Background — The 12-Term MUGE (PAPER_146)

The 12-term MUGE Resonance master equation (§2.2):

$$g_{\text{res}}(r, t) = a_{\text{DPM}} + a_{\text{THZ}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} + U_{\text{g4i}} + a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} + \text{Osc}_{\text{term}} + a_{\text{exp,freq}} + f_{\text{TRZ}}$$

2. The 13th Term — Wormhole Gravitational Acceleration

The Morris-Thorne traversable wormhole metric:

$$ds^2 = -dt^2 + dr^2 + (b^2 + r^2)(d\theta^2 + \sin^2\theta d\phi^2)$$

where b is the **throat radius**. The radial tidal force on matter passing through gives a gravitational acceleration contribution:

$$a_{\text{worm}}(r) = \frac{f_{\text{worm}} \cdot E_{\text{vac,neb}}}{b^2 + r^2}$$

Parameter	Value	Units	Origin
f_{worm}	1.0	—	Wormhole coupling factor

Parameter	Value	Units	Origin
E_vac,neb	7.09×10^{-36}	J/m ³	Nebular vacuum energy density
b	1.0	m	Morris-Thorne throat radius
r	radial distance	m	Distance from throat

At large r: $a_{\text{worm}} \approx E_{\text{vac,neb}}/r^2 \rightarrow$ same $1/r^2$ decay as Newtonian gravity At r=0 (throat): $a_{\text{worm}} = E_{\text{vac,neb}}/b^2 = 7.09 \times 10^{-36} \text{ m/s}^2$

3. 13-Term MUGE — Complete Master Equation

$$g_{\text{res}}^{(13)}(r,t) = \sum_{i=1}^{12} a_i + a_{\text{worm}}(r)$$

Explicitly:

$$g_{\text{res}}^{(13)} = a_{\text{DPM}} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} + U_{\text{g4i}} + a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} + \text{Osc_term} + a_{\text{exp,freq}} + f_{\text{TRZ}} + \frac{f_{\text{worm}} \cdot E_{\text{vac,neb}}}{b^2 + r^2}$$

4. C++ Implementation (from thread 7f9068)

```
double compute_a_wormhole(double r, double b = 1.0,
double f_worm = 1.0,
double E_vac_neb = 7.09e-36) {
return f_worm * E_vac_neb / (b * b + r * r);
}
```

Integrated into compute_resonance_MUGE():

```
double compute_resonance_MUGE(const MUGESystem& sys, const ResonanceParams& res) {
// ... 12 existing terms ...
double a_worm = compute_a_wormhole(sys.r);
return aDPM + aTHz + avac_diff + asuper_freq + aaether_res
+ Ug4i + aquantum_freq + aAether_freq + afluid_freq
+ Osc_term + aexp_freq + fTRZ + a_worm; // 13th term
}
```

5. Physical Magnitude Compared to Other Terms

For r = 1 AU = 1.496×10^{11} m:

$$a_{\text{worm}}(1\text{ AU}) = \frac{7.09 \times 10^{-36}}{1 + (1.496 \times 10^{11})^2} \approx 3.17 \times 10^{-58} \text{ m/s}^2$$

This is 10^{22} times smaller than Newtonian gravity at 1 AU ($5.93 \times 10^{-3} \text{ m/s}^2$), making a_{worm} astrophysically negligible in the Solar System but **dominant** in quantum gravity regimes where $b \sim \ell_{\text{Planck}}$ and $r \sim b$.

6. Distinction from PAPER_153

Feature	PAPER_153 (§2.2)	PAPER_159 (§2.3, this paper)
Equation type	Metric ds^2 (geodesics)	Gravitational acceleration a_{worm}
MUGE integration	Uses fTRZ throat term	Adds explicit 13th acceleration term
CP target	CP3 metric calculator	CP3 resonance MUGE extension
Application	Traversability condition	MUGE resonance sum
Calibrated at $b=1.0$	Yes (throat)	Yes (same $b=1.0$)

7. Python Implementation (CP3)

```
def compute_a_wormhole(r: float, b: float = 1.0,
    f_worm: float = 1.0,
    E_vac_neb: float = 7.09e-36) -> float:
    """
    13th resonance MUGE term: Morris-Thorne wormhole gravitational acceleration.
    a_worm = f_worm * E_vac_neb / (b^2 + r^2)
    """
    return f_worm * E_vac_neb / (b**2 + r**2)
```

Status: ■ Complete | **CP Stage:** CP3 (extend `resonance_muge()`) **Supersedes:** N/A (extends PAPER146) | **Related:** PAPER146 (12-term), PAPER153 (wormhole metric), PAPER091 (resonance framework)